FLEET TEAM DIGEST

737MAX-FTD-29-17002

Issue Title : Hydraulic Fluid Transfer from System B to System A

Airplane Model	ATA	Minor Model(s)
737MAX	2911-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
03/22/2017	06/15/2022	06/24/2021

Estimated Completion Date	Next Update Date

Revision Description

Service documentation has been published.

Applicability

All 737 MAX aircraft.

Description

Several operators have reported cases of chronic hydraulic fluid transfer, primarily from hydraulic system B to hydraulic system A, within their 737 fleets. As a result, the 737 Hydraulic Fluid Transfer Working Together Team (WTT) was formed to investigate and determine the root cause of fluid transfer and evaluate potential mitigating actions.

Background

As stated previously, several operators have reported cases of hydraulic fluid transfer, predominantly from system B to system A, throughout the service life of the 737. Consequently, Boeing formed a working group composed of both Boeing and 737 operator personnel in order to identify the potential pathways of hydraulic fluid transfer. This group performed an analysis of the hydraulic system design and identified components that provide a hydraulic link between systems B and A. The following per-design pathways could, under the right conditions, enable fluid transfer from system to system and are ordered by expected likelihood:

Most Likely:

1) Brake Shuttle Valves

Less Likely:

2) Left Thrust Reverser Shuttle Valve

3) Landing Gear Transfer Valve (LGTV)

4) Rudder Power Control Unit (PCU)

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Eliminated Candidates:

5) Alternate Source Select Valve (ASSV)

6) Elevator Feel Computer

7) Elevator Feel Actuator

8) Accumulator Isolation Valve (AIV)

It is also noted that there are procedural issues that can transfer fluid. Examples are the parking brake set/release procedure, performing gear swings when the airplane is on jacks, and when the left thrust reverser is operated by the standby hydraulic system. These procedural issues are discussed in the Fault Isolation Manual (FIM) section 29-10. This article focuses on non-procedural causes of fluid transfer.

The WTT has collaborated with an airline and has completed a review of raw flight data for 10 different tail numbers and 25 different flight segments. Upon completion of this analysis, Boeing was unable to pinpoint during the course of a single flight fluid transfer from hydraulic system B to A. It was determined that it is difficult to see the transfer over the course of a flight versus a few days and it may be occurring between flights.

In the past several months, we have been working with the WTT lead operator to develop and validate a prototype customized AHM alert which identifies aircraft that are particularly prone to chronic hydraulic fluid transfer. Once this AHM alert identified an aircraft, Boeing asked the WTT lead operator to remove all four brake shuttle valves and replace them with four valves that exhibit zero internal leakage (per SCD and CMM requirements). Boeing then monitored this aircraft's reservoir fluid quantity levels for over one month and found that the chronic fluid transfer phenomenon ceased upon replacing the brake shuttle valves. Boeing also asked the WTT lead operator to send all four removed brake shuttle valves to our hydraulics laboratory for internal leakage testing. This testing verified internal fluid leakage on all four shuttle valves, which more than likely was the cause of the fluid transfer on this aircraft.

A similar analysis was performed on another aircraft in order to verify the root cause of hydraulic fluid transfer. In this case, an additional four shuttle valves were leak tested and internal leakage was confirmed. Root cause of 737 hydraulic fluid transfer has been determined to be high pressure leakage through one or more of the brake shuttle valves. At this time, we have engaged the supplier of the brake shuttle valves (PneuDraulics) and are working with them to coordinate an investigation into the valve build process (as a mitigating measure) and potential corrective action (as a final measure). A decision was made to elevate the project to a Service-Related Problem (SRP) to develop appropriate corrective actions to address this issue.

Status

Service documentation has been published. Reference Service Letter (SL) 737-SL-29-170

Interim Action

To mitigate fluid transfer, Boeing recommends that operators avoid leaving the parking brake set for extended periods (i.e., overnight). As discussed previously, the primary fluid transfer root cause candidate is high pressure internal leakage through the brake shuttle valves, which is most likely to occur during extended parking brake operations. As such, releasing the parking brake for longer parking situations (after chocks are in place) may be helpful.

Boeing would like to note that some operators have reported that they can use the parking brake set/release procedure to purposely move fluid from the overfilled system A reservoir to the under filled system B reservoir without formally servicing the

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reservoirs. This procedure involves turning on and off the Electric Motor-Driven Pumps (EMDP's) in a specific sequence while setting and releasing the parking brake as follows:

1) With wheel chocks in place, turn off both the system A and system B EMDP's.

2) Turn the system A EMDP on.

3) Set the parking brake.

4) Turn the system B EMDP on.

5) Release the parking brake.

6) Turn both the system A and system B EMDP's off.

7) Repeat this procedure if needed.

Boeing would like to note that EMDP's can be overheated if this procedure is used too many times in a short duration. We recommend that EMDP's be operated intermittently a maximum of five times in a five minute period (with a 30 second wait time between each stop and start of the pump). After completing five iterations of the procedure mentioned above, the pumps should either be run continuously for five minutes after the fifth cycle (while monitoring the overheat warning lights) or turn both pumps off and let them cool for more than 30 minutes.

Each iteration of the procedure above will result in 15-20 cubic inches of fluid transfer from system A to system B. As such, the aforementioned procedure is not recommended for transferring larger amounts of fluid between hydraulic systems. Boeing recommends servicing the hydraulic reservoirs when possible.

System pressure fluctuations during single engine taxi can result in small amounts of inadvertent fluid transfer. The general scenario is as follows: Normal hydraulic braking power is provided via hydraulic system B. As long as system B pressure is greater than roughly half the value of system A pressure, the ASSV will remain detented to the "normal brakes" position. During typical aircraft taxi operations (i.e., with both the system B EMDP and system B EDP operating), the hydraulic pumps will have more than enough flow capacity to handle any system demands (i.e., flap/slat movement, flight control actuation, etc.) without a subsequent drop in pressure output. However, if the aircraft was taxiing with the left engine ON and the right engine OFF, this would only leave the ELEC 1 pump (with a limited flow capacity of roughly 6 GPM) to handle the workload of the entire B system. In this case, flap/slat movement (handled exclusively by system B) will result in an immediate system pressure drop in order to maintain the flow required to move these surfaces. Pressure will eventually recover to normal levels; however, if brake application occurs simultaneous to this pressure drop, then then ASSV will shuttle to the "alternate brakes" position and 15-20 cubic inches of system B hydraulic fluid will transfer to system A. Any other system demand that would typically be present during the taxi phase of operation would not generate a high enough flow demand to result in a system pressure drop high enough to prompt shuttling of the ASSV to the alternate position. Lastly, this scenario only holds true when single engine taxiing on the LEFT engine.

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When taxiing on the RIGHT engine only, adequate hydraulic flow and pressure will be available to prevent operation of the ASSV and the need for alternate braking.

Final Action

Boeing plans to implement the new brake shuttle valves in 737 production. Boeing has published Service Letter 737-SL-29-170. PDI has published Vendor Service Bulletin 5059-32-10-62240-1-01.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: 24-MAY-2017 (Complete)

SRP Initiation: 24-MAY-2017 (Complete)

Component-Level Root Cause Established: 26-JAN-2018 (Complete)

Corrective Action Validation Testing Complete: 14-SEP-2018 (Complete)

Solution Selection: 26-SEP-2019 (Complete)

Change Committed: 10-OCT-2019 (Complete)

Service Documentation Available: JUN-2021 (Complete)

Parts Available: JUN-2021 (Complete)

Operator Action

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At this time, Boeing recommends that operators perform 737 MAX IFIM Task 29-10-00-810-803 (for system B to system A transfer). Particular attention should be paid to internal leakage of the brake shuttle valves, which coupled with high rates of leakage to return within the Alternate Anti-Skid Valve, can result in fluid transfer.

Operators should record hydraulic fluid transfer anomalies and maintenance performed. Operators that demonstrate proof of three (3) hydraulic fluid transfer events in a four (4) week period without evidence of external leakage or other related component failure meet the failure criteria and should contact PDI for further action. Hydraulic fluid transfer may be recorded by means of AHM records of reservoir fluid quantities or repeated reservoir rebalancing events entered into logbook data. Alternatively, at the discretion of the customer, the Boeing Fault Isolation Manual (IFIM) procedures may be used to explicitly determine if one or more of the 5059-4 shuttle valve is exhibiting internal leakage.

Lastly, operators should continue to ensure that flight crews follow published hydraulic pump ON/OFF sequencing procedures during aircraft maintenance and operations. Hydraulic pump ON/OFF sequencing can have a significant impact on hydraulic fluid transfer between reservoirs (Boeing believes this is the most likely cause of procedural fluid transfer). As stated above, hydraulic system B provides normal brake power and charge to the brake accumulator. Regardless of dual or single engine taxi, it is ideal to set the parking brake with system B power, which can be done with both system A and B powered (as long as system B pressure is greater than half of system A pressure) or only system B powered. If this occurs, then the parking brake should also be released using system B power as well. This ensures that the system B hydraulic fluid that filled the brakes during parking brake set will be returned to the system B reservoir following parking brake release. If the parking brake was set using system B power and released with only system A powered, then the ASSV would shuttle. This in turn would result in the system B fluid that filled the brake pistons being returned to the system A reservoir (i.e., a net B to A transfer of 15-20 cubic inches of fluid per each ON/OFF sequence).

Please see 737-SL-29-170 for further information.

Supplier Information

PneuDraulics, Inc. (PDI)

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91730

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Reference Categories

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FLEET TEAM DIGEST

Reference Type	Reference Number	Revision Number	Reference Date
Fleet Team Economic (FTE)	FTE-29-14022		05/05/2016
Fault Isolation Manual (FIM)	737MAX IFIM Task 29-10-00-810-803		
Service Letter (SL)	737-SL-29-170		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Brake Shuttle Valve		PneuDraulics	5059-4

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- FLEET TEAM Conference Item : null
- Fleet Team IdeaXchange : Topic of discussion on the Fleet Team IdeaXchange bulletin board
- In Service Occurrences : null

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